PakEmis Sector Technical Note TN03

Fuel Anchoring as Reconciliation by Construction: A Greenhouse-Gas Inventory of Road Transport and Mixed Off-Road Mobile Combustion in Pakistan

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Abstract

Transport inventories are usually built bottom-up from a vehicle fleet and reconciled against fuel supply afterwards, resting their weakest input—fleet activity—under their most-used output. We invert that for Pakistan: road transport (IPCC 1A3b) and mixed off-road machinery are anchored on the road customer class of OCAC's sectoral petroleum sales for FY2023–24 (7209 kt petrol, 5551 kt diesel; those customer classes sum to the national total exactly, excluding bulk industrial, utility, government and railway diesel by construction) and on metered transport CNG. The fleet model distributes this fuel and cannot size it: a bottom-up build on the same data would miss diesel by half and overstate CNG threefold. Class shares are vehicle-kilometres weighted by declared energy intensities and normalised within each fuel, so that judgement moves composition and never magnitude—correcting it from distance-only weighting moved two-wheelers from 42.3 to 34.1 % of on-road CO_2 (13.40 Mt) while the sector total moved by 0.05 Mt. We report 40.19 Mt CO_2e as the *conditional road residual under the current claimant assignment* and 1.73 Mt for mixed off-road machinery, with a two-tier ledger that closes both controls exactly and carries the unresolved retail-channel claimants as status rows, priced at 2.5–10% of the road row and treated as transfers to other IPCC categories rather than reductions. An independent railway cross-check agrees to 3.8 %, and against a fiscal-year-blended EDGAR benchmark the estimate is 0.85 of it—a scope-approximate diagnostic, since EDGAR's transport boundary is inferred rather than published. The reported interval is ±11.4 %, conditional on the boundary; a ten-year fuel-based series and per-fuel provincial allocations follow at no further modelling cost.

Keywords: road transport emissions, fuel-based inventory, off-road machinery, vehicle fleet model, emission factor tables, control total, air pollutants, IPCC 2006 Guidelines, Pakistan, MRV

PakEmis Sector Technical Note TN03. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

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Commit	`6d8e8669a460`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`e6877cc36bb6604c` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

There are two ways to build a road transport inventory and they fail differently.

The bottom-up approach estimates the vehicle fleet, assigns annual kilometres and fuel economy by class, and sums. It produces a rich result—emissions by vehicle type, useful for pro-

jecting the effect of a fleet policy—and a fuel demand that will not match the national petroleum supply. The compiler then adjusts something until it does.

The fuel-anchored approach starts from the supply figure and distributes it across classes. It produces a total that matches national statistics by construction and a composition that is only as good as the fleet model. Nothing needs reconciling because nothing was independently estimated.

Which is better depends on which number is more trustworthy. In Pakistan that is not close: the petroleum supply figure is a metered, taxed, nationally reported quantity, while the vehicle fleet is a registration database that counts vehicles administratively rather than by where they operate, includes a residual catch-all class, and says nothing about how far each is driven. This paper quantifies the gap directly: the bottom-up fuel demand the same fleet data imply, under declared engineering intensity bands, misses the observed diesel control by half and overstates metered CNG threefold (Section 6)—errors in both directions, which is the strongest form of the case for letting fuel set the magnitude.

1.1. What this paper does

We report Pakistan's road transport and off-road mobile machinery for FY2023–24 on the fuel-anchored basis, and treat the consequences of that choice as the subject rather than as an implementation detail.

The consequence that matters most is that anchoring makes a boundary problem explicit. Pakistan's petroleum statistics do publish a sectoral split—OCAC's customer-class sales tables, which this edition anchors on directly—but a customer class is a sales channel, not an end use: the retail Road row still contains diesel bought at pumps for stationary agricultural plant and captive generators. This paper reads the observed customer-class partition into a two-tier fuel-claimant ledger, carves out what a sibling sector measures (household-generator petrol), partitions off-road machinery, cross-checks the rail boundary against an independent source, and states plainly what remains inside the road residual and in which direction it biases the result.

1.2. Why Pakistan, and why anchoring in particular

Pakistan's most recent full national inventory submission reports for FY2017–18 (Ministry of Climate Change, Government of Pakistan, 2022), and the international products that fill the gap must be compared with the years matched, because Pakistan's fuel demand moved sharply after 2022. On a scope inferred from EDGAR's category-resolved Pakistani source files—road, rail and civil aviation—this inventory's corresponding construction is 0.85 of EDGAR's fiscal-year blend of 48.13 Mt (Crippa et al., 2025) (Section 6), sitting *under* the benchmark for the reasons the customer-class boundary predicts. The blend is half of EDGAR's 2023 column and half of its 2024 one, because this sector's July–June reference period falls across both; the comparison is a scope-approximate diagnostic rather than a validation, because EDGAR's transport boundary is inferred rather than published.

Anchoring is particularly appropriate here for a reason specific to the country's fleet. Two-wheelers are the largest single class in Pakistan's road transport—under the reference allocation they receive 13.40 Mt, 34.1 % of on-road CO_2—and their annual kilometres are among the least well-constrained inputs in this Pakistani fleet model. A bottom-up estimate would rest a very large term on one of its weakest inputs. Anchored, that same uncertainty affects only how the fuel is split between classes, which is exactly what happened when this edition corrected the split: two-wheelers fell by six megatonnes, heavy diesel rose past them, and the sector total moved by 0.05 Mt (Section 7).

1.3. Contributions

1. **A fuel-anchored road transport inventory for Pakistan** anchored on the customer-class rows of the national sectoral sales table, so the sector total reconciles with petroleum statistics by construction and the bulk non-road claimants are excluded by the source's own structure (Section 4).
2. **A two-tier fuel-claimant ledger, mostly observed**: the national tier is OCAC's own exhaustive customer-class partition; the road tier carries one relayed carve-out (generator petrol, closed against the residential module), one partition (off-road machinery), an independent rail cross-check between two unrelated sources, and the unresolved retail claimants as bounded status rows instead of silence (Table 2).
3. **A quantified instance of the IPCC parallel-table trap**, in which the plausible wrong table leaves CO_2 exactly right and moves the category by 8 % through N_2O alone—invisible to every CO_2-dominated comparison available (Section 5).
4. **A source-traceable road air-pollutant estimate** using EMEP/EEA Tier 1 fuel-based exhaust factors, which the guidebook derives from its hot and cold Tier 3 method for a typical 2010 EU-27 fleet (no Pakistan-specific cold-start treatment is applied), separate petrol evaporation and tyre/brake/road-surface wear. Transferred fleet factors, technology, sulfur compliance and resuspended road dust retain their actual limitations (Section Appendix D).
5. **A provincial petroleum allocation from OCAC's monthly area editions**, allocated per fuel, with a published diagnostic demonstrating the failure of the registration proxy it replaces and an allocation crosswalk stating which streams ride which proxy (Section 7).

Method shared with the other nine sectors of the national inventory—the module contract, the accounting closures, the parameter-uncertainty treatment—is documented separately (Ahmad, 2026) and not repeated here.

2. Related Work

2.1. Fuel-based and fleet-based transport inventories

The IPCC 2006 Guidelines describe both approaches for 1A3b and are explicit that the fuel-based Tier 1 is preferred

where national fuel statistics are reliable and fleet data are not (IPCC, 2006). That guidance is often inverted in practice, because a fleet-based estimate produces the disaggregation that transport policy wants and a fuel-based one does not.

The position taken here is that the two are not alternatives at the same level. The fuel figure determines the total; the fleet model determines the split. Running the fleet model *inside* the fuel constraint gives the policy-relevant disaggregation without letting it set the magnitude, which is the arrangement the guidelines' preference implies but which is rarely stated as such.

2.2. Global products and their transport scope

EDGAR (Crippa et al., 2025, 2020) reports Pakistani transport across all modes—road, rail, domestic aviation and navigation together. Comparing that against a road-only estimate is a scope error that will make any road inventory look low; Section 6 walks the scope component by component instead, and labels the result honestly as fiscal-year blended and partially scope-harmonized.

Climate TRACE (Climate TRACE, 2026) constructs road transport from asset and activity inference. Both are global models applying globally consistent assumptions, which is their strength across countries and their limitation within one: neither can know that Pakistan's national diesel figure includes agricultural and captive-generation fuel, which is the single largest scope question in this sector.

2.3. Off-road machinery, and where it belongs

IPCC reports off-road mobile machinery outside road transport, and not all under one category: agricultural tractors and plant go to 1A4c ii, while construction and industrial machinery belongs under 1A2 (manufacturing and construction, gvii). This is easy to state and easy to get wrong in a country where the diesel that runs all of it comes from the same retail pumps as the diesel that runs the trucks and is not separately reported—and this inventory gets it only partly right: its off-road class inherits a residual bucket that mixes both, and is reported under the mixture-declaring name *Mixed off-road (1A4c/1A2)* with the misassignment declared in the completeness table and priced as a reporting-recode scenario rather than resolved (Section 9).

The usual treatments are to leave it inside road transport, which overstates 1A3b and leaves 1A4c empty, or to estimate it separately from a machinery population, which double-counts the fuel unless the road control total is reduced correspondingly. Section 4 describes a third: carve it out of the same control total by vehicle-kilometre share, so the two categories partition one quantity of diesel.

2.4. The parallel-table problem

Chapter 3 of Volume 2 publishes separate default tables for road, rail and off-road mobile sources, and the same diesel appears in all of them with identical CO_2 and sharply different N_2O. The distinction reflects real combustion differences—catalyst-equipped road engines against uncontrolled off-road ones—but the tables are easy to conflate because the fuel is bought from the same pump.

The national manuscript documents four instances of this trap across the inventory (Ahmad, 2026). Section 5 treats the one that arises here, and reports the diagnostic the engine emits on every run so the consequence is a number rather than a caution.

3. Study Area and Data

3.1. The control totals

The petroleum activity data are the *Road* customer-class rows of OCAC's Sectoral Sales of Energy Products for FY2023–24 (Oil Companies Advisory Council, 2026)—the sector's reference period under the same July–June fiscal convention the CNG stream pins by test. OCAC publishes this table as sector rows (Domestic, Industry, Agriculture, Transport—itself split into Road, Railways, Domestic Bunkers and two aviation rows—Power, Government) by product columns in tonnes, and the customer classes sum to the printed national total exactly. Road is a *customer class*: retail pump sales. Bulk industrial, utility, government and railway diesel never enters it, so those claimants are excluded from the road control by construction rather than carved out of it.

Fuel	**TJ**	**Share (%)**
Petrol	316 882	57.0
Diesel	238 712	43.0
Total	**555 594**	**100.0**

Table 1: The petroleum road controls for FY2023–24: the OCAC Road rows (petrol = MS + HOBC, 7209 kt; diesel 5551 kt) converted at IPCC net calorific values. CNG is a third, separately controlled activity basis (Section 4) and does not appear here because it is not petroleum. The full claimant structure over these controls is Table 2.

3.2. A provenance audit: the superseded control

The previous edition anchored on national supply figures derived from the Economic Survey's provisional nine-month FY2025 data. Those stored values—10 162 and 8820 kt—were arithmetically inconsistent with their own source note: they equal the nine-month figures times 16/9, not the documented 12/9 annualisation, an annualisation applied twice. The diesel figure coincidentally landed within one percent of the FY2021–22 *national* internal HSD total, so the matched-year EDGAR comparison looked right on a wrong-provenance number; the petrol figure overshot the fiscal-year national total by eleven percent. A match against a benchmark is not provenance. The superseded pair is retained as a named, executable boundary scenario (`legacy_pes_control`, Section 7) rather than deleted—the same discipline the residential sector applies to its superseded dung basis—so the bridge from previously published numbers stays open and honest.

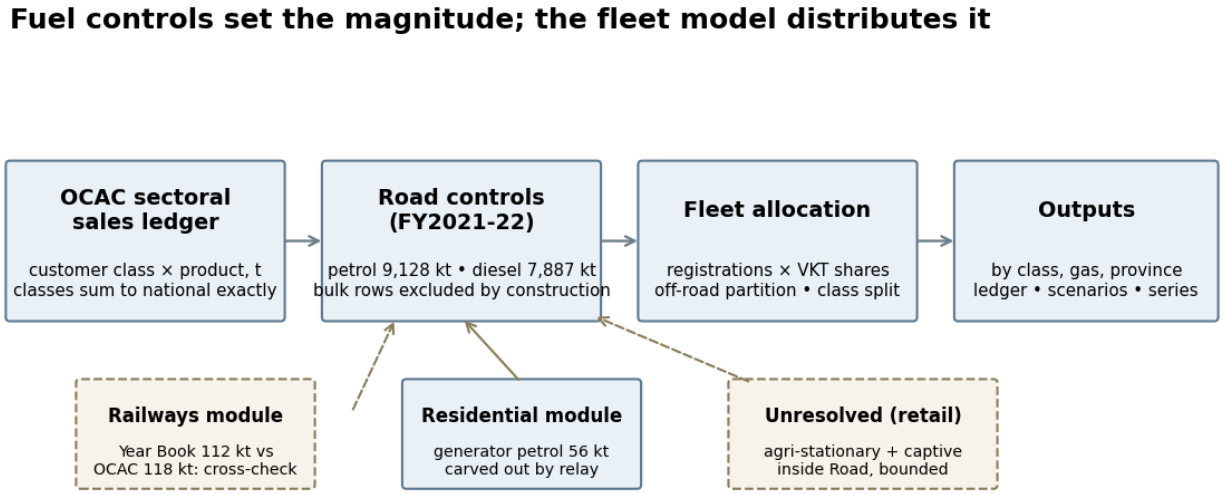


Figure 1: The calculation, end to end: the OCAC customer-class ledger supplies the road controls, the fleet model distributes them, and the cross-sector relays (railways as a cross-check, residential as a carve-out) and the unresolved retail claimants are explicit boxes rather than prose. The high petrol share of the controls is consistent with Pakistan's large two-wheeler fleet.

3.3. What the road rows still include that they should not

A customer class cannot see end uses. Retail pumps sell diesel to trucks, but also to tube-well and thresher owners (stationary agriculture, IPCC 1A4c i) and to captive generator operators (1A2); retail petrol includes fuel carried home for household generators (1A4b). One of these three is carved out—the generator petrol, relayed from the residential module (Section 4)—and the two diesel claimants remain inside, unresolved, biasing the road residual high conditional on the other deductions. Two observations frame them: OCAC books *zero* bulk HSD to its Agriculture row in all ten editions, so the agricultural claim is retail-channel only; and the road share of national diesel spans 0.83–0.91 across the ten-year panel—a customer-class boundary sensitivity that bounds the sales channel, not the contamination within it, which is priced separately by the retail non-road scenarios of Section 7. The complete claimant structure—observed bulk rows, carve-outs, residuals and unresolved status rows—is the two-tier fuel ledger of Table 2.

3.4. The fleet model

Vehicle classes and their shares come from the national registration statistics (Provincial Bureaux of Statistics, 2024), which give counts by class and province. Annual vehicle-kilometres and fuel-split shares by class come from the transport parameter table, whose survival parameters derive from a district-level Punjab transport manuscript (Naveed, 2024) that adopts the survival functions of Guttikunda (2024).

The classes are two- and three-wheelers, four-wheelers in three size bands, light and heavy duty vehicles, buses, and an off-road class. That last one is a genuine complication rather than a clean category, and Section 9 treats it: the source workbook maps its catch-all "other vehicles" entry into off-road wholesale, so the class is a residual as well as a machinery category, and it is reported under the mixture-declaring name *Mixed off-road (1A4c/1A2)*.

3.5. Air-pollutant factors

The air-pollutant calculation (Section Appendix D) uses the held EMEP/EEA 2023 source chapters and their fuel/category/process factors. Exhaust uses OCAC-constrained fuel with Tier 1 factors that average hot and cold operation for a typical 2010 EU-27 fleet, without a Pakistan-specific cold-start treatment; petrol evaporation uses active vehicle-days; tyre/brake and road-surface wear use modelled vehicle-kilometres. The class map explicitly declares CNG-to-LPG chemistry and petrol-bus substitutions. Factor tables preserve source values, applicability and units. NO_x is NO_2-equivalent and NMVOC excludes methane. Policy-based sulfur specifications are conditional regimes rather than measurements of universal fuel compliance.

3.6. What this inventory had not read

Pakistan publishes no nationally representative vehicle-kilometre survey that this study could identify, and no on-road fuel-economy measurement was found. Everything else this sector once described as unobtained turned out to be a reading failure, twice.

Compressed natural gas was the first instance: a substantial CNG fleet whose fuel sits in the gas statistics, absent from the sector entirely while the transport row of OGRA's sector-wise gas consumption table—the same table this inventory already read for the residential control total—sat unread (Oil and Gas Regulatory Authority, 2024, 2026). That row became the sector's third fuel control, and the previous edition drew the lesson: not absent, unread.

The OCAC sectoral tables are the second instance, at larger stakes, and the lesson applied to the sentence that drew it. Two

editions of this paper described OCAC's product-by-sector sales as published but not obtained here. They were in the repository throughout—forty-eight monthly editions carrying CUMULATIVE national tables, fetched by the aviation sector for their jet-fuel columns, indexed, hashed, and already read by the residential sector for a kerosene cell; a distinct population from the thirty-six DIRECT single-month editions below, as the edition inventory (`outputs/ocac_edition_inventory.csv`) now distinguishes. This edition anchors on them (Section 4) and builds its provincial allocation from area pages (Section 7). There is now a third episode of the same lesson: an earlier edition borrowed FY2024–25's provincial shape on the claim that the reference year's area pages were never published, while the thirty-six direct monthly editions carrying them sat fetched and hash-pinned in the repository. All three episodes are kept here, in the data section, as the clearest available statement of what "not obtained" usually means in practice.

4. Methods

4.1. Anchoring: the fleet distributes, it does not size

Emissions for gas g from class c burning fuel f are

$$E_{g,c,f} \;=\; A_f \cdot s_{c,f} \cdot EF_{g,f}, \tag{1}$$

where A_f is the road-class supply of fuel f in terajoules, $EF_{g,f}$ the IPCC default factor, and $s_{c,f}$ the share of that fuel attributed to class c. The shares are normalised energy products,

$$s_{c,f} \;=\; \frac{N_c\, k_c\, q_{c,f}\, e_{c,f}}{\sum_{c'} N_{c'} k_{c'} q_{c',f} e_{c',f}}, \tag{2}$$

over vehicle counts N, annual kilometres k, the fraction q of a class's travel on fuel f, and the per-kilometre energy intensity e in megajoules. The intensities are *declared engineering judgement*—generic band midpoints recorded per class and fuel in the parameter table, not Pakistani measurements, and the same values the fleet-implied fuel diagnostic of Section 6.3 uses.

Because the shares are normalised, $\sum_c s_{c,f} = 1$ by construction, and therefore $\sum_c E_{g,c,f} = A_f \cdot EF_{g,f}$ regardless of what the fleet model says. The sector total is a function of the fuel controls and the IPCC factors alone.

That normalisation is also what bounds the declared intensities. No choice of e can move a control total or the sector magnitude; it moves *composition*, and—through the off-road partition, which is one cell of the same normalisation—which IPCC category carries a fixed quantity of diesel. Both consequences are priced as scenarios in Section 7 rather than folded into an interval. Admitting a judgement into the calculation at all is a real cost, and it is paid where it does least damage: a quantity nobody has measured determines which class is said to have burnt the fuel, never how much fuel there was.

The previous edition of this paper printed the equation above with e in it while the code applied $e \equiv 1$—normalising distance alone, which asserts that a motorcycle-kilometre burns as much petrol as a car-kilometre and a van-kilometre as much diesel as a truck-kilometre. Neither is true, and the correction is not small: the two-wheeler share of the petrol control falls from 0.8119 on distance alone to 0.6100 and the class that dominates on-road CO_2 changes (Section 7). The equation and the engine now agree, and a test recomputes both normalisations from the parameter files to assert which one the published weights came from.

This is the property that makes the fleet model's weaknesses tolerable. An error in two-wheeler annual kilometres moves fuel between two-wheelers and everything else; it does not change how much fuel the road class bought. The estimate that policy uses for the national total is therefore insulated from the input that a bottom-up inventory would rest it on.

Unless stated otherwise, every ± figure in this paper and its supplement is an approximate 95 % relative half-width: IPCC Approach 1 propagation where a value is quoted for a category or the sector, and the 2.5th–97.5th percentile band of the Monte Carlo where a result is labelled as such.

4.2. The fuel ledger: two carve-outs' worth of structure, mostly observed

The previous edition partitioned a national supply figure by argument—carve this out, declare that unresolved. Anchoring on the OCAC customer-class rows turns most of that argument into observation: the national tier of the ledger is OCAC's own partition, whose classes sum to the printed national total exactly, and only the road tier still needs carve-outs and status rows. Table 2 is the complete structure; the engine writes it on every run and aborts if any of the four (fuel, tier) ledgers fails to close.

The ledger closes in mass. What it does not show is the step from mass to the energy the engine actually distributes, and that step is not symmetric between the two fuels: Table 3 makes it explicit.

The claimant list is exhaustive by the structure of the source, not by enumeration: OCAC's TOTAL INTERNAL row closes over its sector rows, so every tonne sold domestically sits in exactly one class. Modes and users a reviewer might ask after are visible rows, not omissions—domestic navigation is the Domestic Bunkers row (3.1 kt, implying about 0.010 Mt CO_2e on IPCC water-borne factors, screened per held edition in `outputs/transport_navigation_screen_2024.csv`, where the FY2023–24 row is less than half this size), military and government fuel is the Government class, mining and bulk captive diesel are inside Industry.

4.3. Two relayed carve-outs, one partition, and a rail cross-check

4.3.1. Household generator petrol, closed against the residential module

The petrol Road row counts every litre sold at the pump, including fuel carried home in cans for small household generators—stationary combustion that IPCC books to 1A4b, not road transport. The residential sector derives that mass from HIES 2024–25 generator-fuel expenditure (Pakistan

Fuel	Tier	Claimant	IPCC	kt	Share
Diesel	national	National internal HSD sales (opening)		6 234.2	1.000
Diesel	national	Domestic	1A4b	0.0	0.000
Diesel	national	Agriculture (bulk)	1A4c i	0.0	0.000
Diesel	national	Industry (incl. bulk captive)	1A2	400.4	0.064
Diesel	national	Railways	1A3c	107.9	0.017
Diesel	national	Domestic bunkers (navigation)	1A3d	3.1	0.000
Diesel	national	Power (utility)	1A1a	41.1	0.007
Diesel	national	Government	various	130.3	0.021
Diesel	national	**Road (retail pump sales)**	1A3b	**5 551.4**	0.890
Diesel	road	Off-road machinery partition	1A4c ii/1A2 gvii	492.5	0.089
Diesel	road	**On-road residual**	1A3b	**5 059.0**	0.911
Diesel	road	Stationary agricultural plant (retail)	1A4c i	unresolved here	
Diesel	road	Captive generation (retail)	1A2	unresolved here	
Diesel	road	Other retail stationary/non-road uses	various	unresolved here	
Petrol	national	National internal MS+HOBC sales (opening)		7 252.0	1.000
Petrol	national	Industry	1A2	25.7	0.004
Petrol	national	Government	various	16.9	0.002
Petrol	national	**Road (retail pump sales)**	1A3b	**7 209.4**	0.994
Petrol	road	Household generators (carve-out)	1A4b	56.3	0.008
Petrol	road	**On-road residual**	1A3b	**7 153.1**	0.992

Table 2: The two-tier fuel-claimant ledger, FY2023–24 (`outputs/transport_fuel_ledger_2024.csv`). National-tier rows are OCAC's own customer classes—observed, not argued—and close to the printed national totals exactly. Road-tier unresolved rows carry no quantity by design: the on-road residual is biased high by them, conditional on the other deductions. The third unresolved row exists because the named claimants are not proven exhaustive within pump sales; the retail non-road scenarios of Table 10 price the whole within-channel claim at 2.5–10%. The ten-year road share of national diesel (0.83–0.91) is a customer-class boundary sensitivity—it bounds the sales channel, not the contamination within it. Closure is an arithmetic property of the ledger, not evidence that the residual is road; the evidence for the residual's composition is the customer-class semantics of the Road row, and its remaining contamination is these rows, never silence.

Fuel	Step	kt	TJ
Petrol	OCAC Road row (MS+HOBC)	7 209.4	319 377
Petrol	less household generators	−56.3	−2 495
Petrol	**operative control**	**7 153.1**	**316 882**
Diesel	OCAC Road row (HSD)	5 551.4	238 712
Diesel	less off-road partition	−492.5	−21 177
Diesel	**operative on-road control**	**5 059.0**	**217 535**
CNG	OGRA metered gas (gross)	—	23 876
CNG	× gross-to-net 0.90	—	−2 388
CNG	**operative control**	—	**21 488**

Table 3: The mass-to-energy bridge (`outputs/transport_fuel_bridge_2024.csv`). The two petroleum figures in Table 1 are not the same kind of number, and this table is the label they were missing: the petrol energy is *net* of the generator carve-out, which is subtracted in mass before the calorific conversion, while the diesel energy is *gross* of the off-road partition applied downstream of it. A reader multiplying the printed 7209 kt by 44.3 TJ Gg^{-1} gets 319 377, not 316 882; both are correct, for different quantities. CNG appears here and not in the claimant ledger because it claims no share of either petroleum control, yet 21 488 TJ of it sits outside a fuel table whose two rows sum to 100 %.

Bureau of Statistics, 2025) (119 responding households) over a declared administered petrol price of 270 Rs/litre (Government of Pakistan, 2025)—the mid-band value for the survey field year, whose gazette-notified range ran roughly Rs 249–293: 56 kt, 0.6 % of the petrol control. Because the survey year is 2024–25 and the control year FY2023–24, the carve-out is a *cross-year proxy*, declared as such and small enough that the mismatch is immaterial. It is subtracted through a relayed accessor—computed once, in the sector that reports it—and a test asserts the equality with ==.

4.3.2. *Off-road machinery, carved by share rather than estimated*

Tractors, agricultural plant and—because the source class is also a residual bucket—construction machinery burn diesel from the same retail pumps, so their fuel is inside the Road row and must be partitioned out. The fleet model normalises its shares over *all* classes including the off-road class, deliberately: renormalising after dropping off-road would give the on-road classes 100 % of the diesel, and any separate off-road estimate would then claim its share a second time. Keeping off-road inside the normalisation means the two categories partition one quantity of diesel exactly, and the memo item that previously carried off-road as a CO_2-only note is asserted to be empty, so the fuel cannot appear in both places. The category is reported under the mixture-declaring name *Mixed off-road (1A4c/1A2)* (Section 9).

The partition share is one cell of that normalisation—the off-road class's energy weight within diesel—read from it rather than recomputed, so conservation of the diesel control is structural rather than a property two calculations happen to share. On the energy weighting the share is 0.0887, against 0.0999 on distance alone: off-road machinery burns *less* per kilometre than the on-road diesel fleet average—7 MJ km^{-1} against roughly 9.5, because buses and heavy goods vehicles dominate that average—so weighting by energy shrinks the off-road share and moves diesel back to the road category. That is a 62 kt reclassification between two IPCC categories, not a change in how much diesel Pakistan burned.

4.3.3. *The survival model, and where it stops operating*

Registration tables report cumulative stock, so the fleet model differences them into entering cohorts,

$d(y_0, c) = \max\{0, N(y_0, c) - N(y_0 - 1, c)\}$, and survives each cohort with a Weibull function $S(t) = \exp\left[-(t/\lambda)^{\beta}\right]$ whose scale is set from the class's median service life, $\lambda = L/(\ln 2)^{1/\beta}$, so that $S(L) = 0.5$ exactly. The active fleet in year y is $F(y, c) = \sum_{y_0 \le y} d(y_0, c)\, S(y - y_0)$, evaluated per district and class.

The $\max\{0, \cdot\}$ discards reported *decreases*, which the source series contain—deregistrations, boundary changes, and a documented change of reporting basis in one province—so the survived cohorts can exceed the stock the same table reports. At district-and-class level the uncapped ratio reaches 402. The model therefore caps the active fleet at reported stock cell by cell and counts how often the cap binds, rather than inheriting the defect.

It binds often: 60 of 115 four-wheeler cells, 69 of 116 light-duty, 68 of 114 heavy-duty, 59 of 114 off-road, and 34 of 115 two-wheeler cells. The consequence is worth stating plainly rather than leaving in a column of the supplement—for the capped-heavy classes the ratio of active to registered vehicles sits at 0.97–0.99, so the Weibull is largely inoperative there and the class split is closer to a registration proxy than the survival machinery suggests. Under anchoring that is a distributional weakness, not a level error, but it is a weakness, and it is one more reason the class results are reported as a bracketed range rather than a number.

4.3.4. Railways: from deduction to independent cross-check

The previous edition subtracted the railway operator's metered purchase (Pakistan Railways, 2024) from a national diesel figure. Under the customer-class controls no subtraction is needed—Railways is its own OCAC row—and the relay becomes something better: an *independent cross-check* on the boundary. Year-matched, the Year Book's FY2023–24 figure (132,588 thousand litres = 112.0 kt at 0.845 kg/l) and OCAC's sold 107.9 kt agree to 3.8 %; the FY2021–22 historical row (142,086 thousand litres = 120.1 kt; Year Book 2022–23 Principal Statistics) gives the conservative cross-year comparison at 11.2 %. A regression test confirms the two independently sourced figures have not accidentally collapsed to one stored value; their independence rests on their separate reporting chains—metered purchases against member-supplied sales. Those observed discrepancies anchor the controls' declared 10 % uncertainty.

4.4. CNG: a third control on its own activity basis

Compressed natural gas never touches the petroleum controls. Its activity is the FY2023–24 row of the transport series in OGRA's State of the Regulated Petroleum Industry 2023–24, Tables 3.31 and 6.1 (Oil and Gas Regulatory Authority, 2024)—62 MMCFD as an annual average, metered utility gas reported in two tables of the same edition that agree—converted from volume at a gross heating value of 1000 Btu/scf (Sui Northern Gas Pipelines Limited, 2024; Sui Southern Gas Company Limited, 2024) and a gross-to-net factor of 0.90 (IPCC factors are defined on a net basis; the utilities bill gross), then run through the same IPCC road factors. The heating value is the declared assumption in the stream and Section 7 publishes the 900 Btu/scf case beside it. The fleet model's CNG vehicle-kilometres are *not* used to size the stream; they set its geography and its air-pollutant share, and Section 6.5 reads the two bases against each other as a diagnostic.

4.5. What remains inside, and in which direction

Retail-bought diesel for stationary agricultural plant—tube wells and threshers—and for captive generation stays inside the Road row, and this inventory has not separated it. It biases the road residual high, conditional on the other deductions being right; we state the direction rather than apply a correction, because a guessed subtraction would move the sector total on the basis of a number nobody has measured. The residual is bounded rather than merely declared: bulk agricultural HSD is observed zero in all ten OCAC editions, and the boundary scenarios of Section 7 span the ten-year road-share envelope. Captive gas self-generation is estimated elsewhere in this inventory from metered captive *gas* (Ahmad, 2026); its retail diesel component is not, and the two are not interchangeable.

5. Emission Factors, and a Trap That Leaves CO_2 Correct

5.1. Which table, and why it matters here

IPCC Volume 2 Chapter 3 publishes separate default tables for road, rail and off-road mobile sources. Across them the CO_2 factor for a given fuel is identical, because CO_2 depends only on the carbon in the fuel and on complete oxidation. The CH_4 and N_2O factors are not.

Category	Table	CO_2	N_2O
		kg/TJ	
Road diesel	3.2.2	74 100	3.9
Off-road machinery	3.3.1	74 100	28.6
Railways	3.4.1	74 100	28.6

Table 4: The same diesel in three tables. CO_2 is identical by construction; N_2O differs by a factor of seven, because off-road and locomotive engines generally carry no catalyst and run lean at high load, which is the N_2O-forming regime.

5.2. What the wrong table costs, quantified

Off-road machinery must take Table 3.3.1. If it took the road table instead, the engine reports the diagnostic on every run: off-road N_2O would be 0.022 Mt instead of 0.161 Mt, and the category total 1.59 Mt instead of 1.73 Mt—8 % low.

The reason this is worth a section rather than a footnote is the failure signature. CO_2 is 91 % of the off-road category and is *identical* between the two tables, so an inventory making this error would still reconcile against the national diesel supply, still agree with EDGAR to within its usual margin, and still pass every mass-balance check the compiler had. Nothing available as a validation would catch it. The only thing that catches it is reading the guidelines carefully enough to notice that there is more than one table— precisely the step that time pressure removes.

It is also directional. Both the road and energy-industry tables carry the *lower* trace-gas values and are the more familiar ones, so a community making this error at random rates would be biased low on non-CO_2 transport emissions rather than scattered around the truth.

A regression test pins the factor, so substituting the road value fails the build.

5.3. Fossil methane, and why it is flagged

Road and off-road diesel and petrol are fossil fuels, so their CH_4 takes the AR5 fossil GWP of 30 rather than the biogenic 28. Every methane-emitting component in this inventory declares which it is, and a contract test asserts both that the declaration is present and that the CO_2e conversion used the corresponding value.

The difference is small here—methane is 0.40 Mt CO_2e across the sector—but the flag is not optional, because the same field distinguishes dung-cake methane from natural-gas methane in the residential sector where it matters considerably more.

5.4. No separate oxidation factor

The 2006 Guidelines assume complete oxidation and fold it into the emission factor; Table 1.4 carries an explicit oxidation column of 1.0 for every fuel. Applying the 1996-vintage 0.98 on top would double-discount. This inventory's waste sector *does* carry an oxidation factor, because open burning genuinely is incomplete, and a test pins both conventions so neither can be "corrected" into the other.

6. Validation

6.1. Against EDGAR: fiscal-year blended, category-mapped

Matching the period comes first, because it moves the comparison more than anything else in this section—and it cannot be done exactly. EDGAR reports calendar years; this sector reports July 2023 to June 2024, so half of its reference period falls in each of EDGAR's 2023 and 2024 columns. The closest annual approximation EDGAR's own data supports is the half-and-half blend, $0.5 \times 47.50 + 0.5 \times 48.76 = 48.13$ Mt, and that is what this paper compares against. The single-year comparisons are kept beside it, because the spread between them *is* the temporal uncertainty and should be visible rather than chosen. Getting the period right has history here: one earlier edition compared a fiscal year against a single calendar column and called it year-matched, and the rc4 edition led with a 0.70 ratio whose engine side mixed the harmonised FY2023–24 road figure with pre-harmonisation FY2021–22 rail and aviation rows against a CY2021/CY2022 EDGAR blend—three vintages in one number, labelled "all FY2021–22". That construction is reproduced at rank *superseded* in the comparison table so it remains checkable, and it must not be quoted as current.

Matching the scope comes second, and this revision does it from EDGAR's own source categories rather than by assertion—though the evidence is inferred rather than published. The GHG booklet defines Transport only as "mobile combustion (road & rail & ship & aviation)", with no IPCC split at all. EDGAR's category-resolved *air-pollutant* files do carry the Pakistani rows explicitly: road (1.A.3.b), railways (1.A.3.c) and civil aviation (1.A.3.a) are present, there is *no* water-borne navigation row (1.A.3.d) for Pakistan at all, and other transportation (1.A.3.e) is nil (`data/activity/transport/edgar_pak_transport_categories.csv`). Those files are the v8.1 air-pollutant release, not the 2025 GHG booklet the benchmark value comes from, so the scope is *inferred from EDGAR's category-resolved Pakistani source files* rather than documented by the product being compared against. The distinction is not pedantry—it is the same class of claim this paper elsewhere insists on labelling—and it would be retired by EDGAR's per-category GHG product, which this study attempted to fetch and could not obtain. On that inference EDGAR's Pakistani transport is road + rail + aviation, and the matched construction on this side is the same three components.

The headline: road residual (40.19 Mt) plus railways (0.39 Mt strict 1A3c traction, FY2023–24—the non-traction 0.005 Mt is excluded because EDGAR's inferred scope has no such category) plus domestic aviation (0.30 Mt, FY2023–24) gives 40.88 Mt, or 0.85 of EDGAR's fiscal-year blend of 48.13 Mt. Against the 2024 column alone it is 0.84 and against 2023 alone 0.86—the width of that spread is what "fiscal year against calendar year" costs, and it is smaller than the scope choices around it. Including the mixed off-road class—which EDGAR books outside its transport aggregate—gives 42.62 Mt, 0.89. Every construction is tabulated with its rank in `outputs/national_edgar_transport_comparison.csv`, including the superseded rc4 construction above. The engine sits *under* the benchmark in all of them, and the retail-channel contamination that pushes this paper's residual the other way is priced by the retail non-road transfers rather than asserted away. The gap between 0.70 and 0.85 is not an engine change: it is what moving both sides of the ratio onto the same period does, because the CY2021/CY2022 EDGAR window straddled the demand peak that preceded the mid-2022 contraction.

One property of that comparison is worth naming because it is reassuring rather than convenient: the off-road-inclusive construction is almost unmoved by the class-split reweighting. Energy weighting moves diesel *between* 1A3b and 1A4c, so a total spanning both barely notices, which is what a partition should look like from outside.

One further construction is retained as history. On calendar-year 2022 controls, rebuilt from the monthly panel (January–June direct editions plus the December cumulative, which is July–December year-to-date by the panel's own tested identity), the construction is about 0.82 of EDGAR's 2022 column: calendar 2022 straddles the mid-2022 demand collapse, so that comparison is honest and wide at once (`outputs/transport_calendar_year_controls_2024.csv`; provincial detail is structurally unavailable for the calendar year's second half). It is ranked superseded beside the rc4 blend, for the same reason. And the previous edition sat *above* EDGAR on a wrong-provenance control (Section 3); that reversal, produced by reading the controls from their

Component	This paper	EDGAR "transport"	Period	Treatment in the comparison
Road (1A3b)	conditional residual, 40.19 Mt	1.A.3.b present	FY2023–24	compared
Railways (1A3c)	separate sector, 0.39 Mt	1.A.3.c present	FY2023–24	added
Domestic aviation (1A3a)	separate sector, 0.30 Mt	1.A.3.a present	FY2023–24	added, on the same fiscal year
Domestic navigation (1A3d)	OCAC Bunkers row, 3.1 kt ≈ 0.010 Mt (screened)	NO 1.A.3.d row for Pakistan	FY2023–24	on neither side of the headline; all ten held editions converted in the navigation screen
Mixed off-road	1.73 Mt, reported 1A4c	outside EDGAR transport (1A4/1A2)	FY2023–24	excluded from headline; secondary line includes it
Bulk captive/industrial diesel	excluded by construction (Industry row, 400 kt)	in EDGAR Industrial Combustion	FY2023–24	outside both transport totals
Retail agri-stationary + captive	inside road residual, unresolved	likely 1A4/1A2	FY2023–24	pushes this paper up; priced by the retail non-road transfers
Calendar convention	FY2023–24 (July–June)	calendar 2023 and 2024, blended	—	primary comparison; single years and superseded rebuilds shown below

Table 5: The scope-and-period crosswalk, on EDGAR's own category evidence. Both sides are AR5 CO_2e, all components on the FY2023–24 reference period against the blended CY2023/CY2024 EDGAR columns. Because EDGAR's transport boundary is inferred from a different release rather than published with the benchmark, every row here is a scope-approximate diagnostic, not a validation.

primary table, is itself evidence that the earlier agreement was coincidence rather than validation.

6.2. *Internal closure checks*

Four properties are asserted on every run and can fail. Closure is an arithmetic property of the ledger, not evidence that the residual is road; the evidence for the residual's composition is the customer-class semantics of the control row, and its remaining contamination is stated in Table 2 rather than hidden by the arithmetic.

The shares sum to one. $\sum_c s_{c,f} = 1$ for each fuel, so the class emissions sum to the control total times the factor. This is what makes the anchoring claim true rather than approximately true.

The off-road memo is empty. The previous treatment carried off-road as a CO_2-only memo item outside all totals. Now that it is a reported category, a test asserts the memo contains nothing, so the same diesel cannot appear as both.

All four ledgers close. Opening equals claims plus closing residual for each (fuel, tier) ledger, to 10^{-9} relative; the run aborts otherwise. The unresolved claimants carry no quantity by construction, so nothing can be invented into the closure.

The rail figures agree without being equal. The year-matched Year Book row and the OCAC Railways row must sit within tolerance of each other and must *not* coincide. The inequality is a regression guard against the two independently sourced fields accidentally collapsing to one stored value; their independence is established by their separate reporting chains, not by the test.

6.3. *The fleet-implied fuel demand, against the observed controls*

The comparison the anchoring argument rests on can now be shown rather than asserted. A conventional bottom-up inventory would compute each fuel's demand from the fleet: $F_f = \sum_c \text{vkt}_{c,f} \times e_{c,f}$. Using this paper's own vehicle-kilometres and *declared engineering bands* for the per-class energy intensities $e_{c,f}$ (judgement values, not Pakistani measurements—the same epistemic class as the CNG check's plausibility band, `outputs/transport_cng_crosscheck_2024.csv`, against the metered series in `outputs/transport_cng_2024.csv`), the bottom-up build misses the observed controls in both directions:

Fuel	Fleet-implied (TJ)	Observed (TJ)	Ratio	Verdict
Petrol	208,714–374,877	316,882	0.84	fleet-model low
Diesel	112,392–205,316	238,712	0.59	fleet-model low
CNG	54,459–98,535	21,488	3.05	fleet-model high

Table 6: Fleet-implied fuel demand at the declared intensity bands (low–high; the ratio is band-midpoint over control) against the observed controls (`outputs/transport_fleet_fuel_diagnostic_2024.csv`). Even the high band undershoots the diesel control, while even the low band overshoots metered CNG: a bottom-up build on this fleet data would err by a factor of two in one direction and three in the other. The diagnostic flags and never adjusts.

This is the demonstration behind the paper's central design choice. Fuel observed at the point of sale sets the magnitude; the fleet model, which cannot even reproduce the fuel, is confined to distributing it.

One caveat on this diagnostic is owed to the reader, because this edition created it. The intensity bands used here are now also the weights of the class split (Section 4), so the diagnostic is no longer independent of the composition it is quoted beside—both read one parameter set, deliberately, so the two cannot silently diverge. What the diagnostic still tests independently is the *magnitude*: the class split is normalised and therefore cannot reproduce a fuel total at all, whatever intensities it is given. That is the comparison in the table, and the reweighting did not change it by a tonne.

6.4. *What is not validated*

The class split has no independent check at all. No nationally representative Pakistani vehicle-kilometre survey has been identified for this study, and no Pakistani on-road fuel-economy measurement either, so the composition in Section 7 rests on registration counts, assumed annual kilometres and declared energy intensities—three inputs, none of them measured in the country. This is stated here rather than in the limitations alone because a reader may reasonably want the class table and should know that it is the least supported thing in the paper. It is also why the class results are reported as a bracketed range: the composition scenarios of Section 7 move the two-wheeler

share of on-road CO_2 across roughly a quarter of its own value without moving the sector total at all.

6.5. *CNG on two activity bases*

CNG is the one fuel in this sector for which two independent descriptions of the same vehicles exist. The emissions come from OGRA's metered gas. The geography and the air-pollutant layer come from the fleet model, which reports CNG vehicle-kilometres by class. Neither is derived from the other, so dividing one by the other tests whether they describe the same fleet—the same move the power sector makes with its implied heat rate.

The metered gas is 21 488 net terajoules—the reported volume converted at a gross heating value of 1000 Btu/scf and then to a net basis at 0.90, the IPCC convention, with both parameters pinned equal to the captive-power category's by test. The fleet model puts 8.55 % of road vehicle-kilometres on CNG, 2.36×10^{10} of them. Together, implying $0.912\,\mathrm{MJ\,km^{-1}}$.

That is low. Published consumption rates put a CNG three-wheeler nearer $1.6\,\mathrm{MJ\,km^{-1}}$ and a CNG car nearer 3, and 74 % of this modelled travel is three-wheelers, so the metered gas is well short of what the modelled distance would require. The two bases disagree, and the direction says the fleet model overstates CNG travel.

Three qualifications. The per-vehicle consumption rates behind that comparison are generic engineering figures rather than Pakistani measurements, so the check flags a disagreement and never adjusts anything. The disagreement is confined to what rides vehicle-kilometres—the provincial split and the air-pollutant layer—because the emissions come from the gas. And it is not a new weakness: this sector already records that its class split rides vehicle-kilometres and over-weights small vehicles, and already argues that magnitude must therefore come from metered fuel. CNG is simply the first category where an independent fuel quantity exists to test that argument, and the test agrees with it.

7. Results

7.1. *The sector*

The conditional road residual—road transport under the current claimant assignment—emitted 40.19 Mt CO_2e in FY2023–24 and mixed off-road machinery 1.73 Mt, together 41.93 Mt on AR5 metrics. The AR6 total is 41.94 Mt: the two differ by 0.02 Mt because the sector is 97.4 % CO_2, whose GWP is unchanged between metrics.

7.2. *The three fuels*

CNG is the smallest of the three and the only one not carved from the petroleum controls. Its 1.28 Mt rests on 62 MMCFD of metered gas, converted at a gross heating value of 1000 Btu/scf and then to the net basis IPCC factors are defined on at 0.90—the gross-to-net convention for natural gas, shared with the captive-power category and pinned equal to it by test. The heating value is the assumption in the stream, as it is for captive power, so the alternative is published rather than buried:

By gas the sector is 40.85 Mt CO_2, 0.67 Mt N_2O and 0.40 Mt CH_4, all as CO_2e. That nitrous oxide is nearly twice the methane is worth noticing: it is a consequence of the off-road factor discussed in Section 5, and an inventory using the road table throughout would report roughly a quarter less N_2O while its CO_2 stayed identical.

Category	**Mt CO_2e (AR5)**	**AR6**	**±%**
Road transport (1A3b)	40.194	40.207	11.8
Mixed off-road (1A4c/1A2)	1.732	1.737	28.2
Total	**41.926**	**41.944**	**11.4**

Table 7: Transport-sector emissions, FY2023–24, on the OCAC Road-row controls. Intervals are parametric and conditional on the claimant boundary; the boundary itself is a structural axis, priced in Table 10. The off-road interval is additionally conditional on the vehicle-kilometre share that carves the class out. The sector band is narrower than either category's for a directional reason: uncertainty in the road/off-road *partition* is negatively coupled between the two categories and cancels when they are recombined, while the control and carbon-factor uncertainties they share are positively correlated and survive—so the sector inherits only the genuinely shared terms.

Fuel	**Mt CO_2e**	**Activity basis**
Petrol	22.542	OCAC Road row, net of generator petrol
Diesel	16.370	OCAC Road row, net of off-road partition
CNG	1.282	OGRA metered sector gas

Table 8: On-road emissions by fuel. CNG comes from a different statistic—gas rather than petroleum—and is the only stream not carved from the Road rows.

Case	**Sector, Mt CO_2e**
CNG at 1000 Btu/scf (reported)	41.926
CNG at 900 Btu/scf	41.798

Table 9: Pakistani pipeline gas runs 900 to 1000 Btu/scf depending on field and RLNG share, and nothing read here pins it. The sector moves by 0.13 Mt across the span—a tenth of the CNG stream.

7.3. *Uncertainty: one reported interval, and its hierarchy*

One interval cannot describe this sector honestly, so the paper reports a hierarchy and never blends its levels.

Reported interval. ±11.4 % (Approach 1, a 95 % relative half-width), *conditional on the current claimant boundary*. This is the number to quote.

Alternative covariance treatments. The sector's grouped treatment—control and calorific value shared within each fuel, the fleet term demoted to distribution-only—gives 7.4 % to 10.1 % depending on whether the two petroleum controls, which come from one table of one publication, share one error.

Cross-sector shared-factor sensitivity. The pooled Monte Carlo (Supplementary Note S1) reports ±12.9 % under module-shared parameters and ±12.9 % with the shared fuel-carbon factors correlated across sectors—the common-random-numbers rebuild made the correlation scenario a no-op for this sector.

The claimant structure, as scenarios. The boundary itself is a discrete choice and is never folded into an interval. Table 10

prices it, and reports each case by category rather than by sector total alone—because for two of these axes the sector total is the least informative number in the row. The off-road partition cases move the sector by a rounding error while moving the off-road category by half its own size, and a reader given only the total would conclude the partition hardly matters.

Case	Road	Off-road	Sector
Reference (OCAC Road rows, FY2023–24)	40.194	1.732	41.926
National HSD minus observed bulk rows	40.203	1.733	41.936
Road-share envelope, low (0.864)	39.709	1.681	41.390
Road-share envelope, high (0.908)	40.507	1.766	42.273
Retail non-road at 2.5% of Road diesel	39.785	1.732	41.517
Retail non-road at 5%	39.375	1.732	41.108
Retail non-road at 10%	38.557	1.732	40.289
Off-road partition at 5%	40.889	0.976	41.866
Off-road partition at 15%	39.093	2.929	42.022
Class intensities at band low	40.499	1.401	41.900
Class intensities at band high	40.027	1.913	41.941
Calendar-2022 controls	47.854	2.147	50.002

Table 10: The boundary and composition axes, structurally (`outputs/transport_uncertainty_scenarios_2024.csv`; all Mt CO_2e AR5). The two defensible constructions agree to 0.04 % because Road plus domestic bunkers *is* the national total minus the observed bulk rows. The road-share envelope is a customer-class boundary sensitivity; the retail non-road rows are the within-channel sensitivity it cannot provide, and are *transfers* rather than reductions—the fuel is rebooked to 1A4c i, and the national effect is only the trace-gas table difference, −0.022 Mt even at 10 %, because CO_2 per terajoule is identical in both tables. The partition and intensity rows leave the sector almost unmoved while redistributing between the two categories: that invariance is the anchoring property, shown rather than claimed. The calendar row is a convention case, not a candidate: CY2022 straddles the mid-2022 demand collapse.

Provenance-audit case	Sector, Mt CO_2e
Superseded PES control (×16/9)	61.524
PES arithmetic corrected, FY2025 vintage	46.326

Table 11: Provenance audit, kept apart from the boundary cases because these are bridges, not candidates: the first reconstructs the previous edition's published basis exactly, the second shows what the FY2025-for-2022 vintage mismatch was hiding (Section 3).

Composition, bracketed rather than declared. The class split still has no independent validation, but it is no longer only a declaration: the declared intensities that weight it have bands, and moving the whole fleet to either end gives the class-intensity rows above. Those rows are a *correlated* perturbation, which the within-fuel normalisation largely absorbs—the two-wheeler share of on-road CO_2 moves by under a percentage point—so reporting them alone would understate the composition uncertainty. The honest envelope comes from moving classes against one another: with two- and three-wheelers at one end of their bands and every heavier class at the other, the two-wheeler share runs several points either side of the reference—the superseded run's envelope spanned twelve points—and the rebuilt cases SHIP as scenario rows of the published table (`class_intensity_low/high` and `intensity_ratio_light_low/high` in `outputs/transport_uncertainty_scenarios_2024.csv`, beside the `class_split_declaration` row). Both ends leave every fuel total exactly where it was. The off-road partition's share is priced by the 5 and 15 percent rows, and the class's agriculture-versus-construction factor mix is *provably invariant*: IPCC Table 3.3.1's Agriculture and Industry diesel rows are numerically identical, an executable scenario confirms the zero, and only the road-table trap (Section 5) moves the answer.

7.4. *Composition, which is what the fleet model determines*

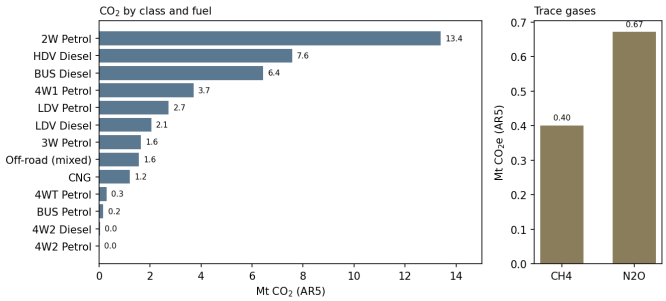


Figure 2: CO_2 by class and fuel, with the trace gases beside on their own scale.

Class and fuel	Mt CO_2
Two-wheeler petrol	13.395
Heavy-duty diesel	7.578
Bus diesel	6.441
Four-wheeler (small) petrol	3.711
Light-duty petrol	2.732
Light-duty diesel	2.059
Off-road machinery diesel	1.569
Three-wheeler petrol	1.641
CNG (all classes)	1.205
Remaining classes	0.522

Table 12: CO_2 by class, on the energy-weighted allocation. Two-wheelers receive 34.1 % of on-road CO_2—still the largest single class, in a fleet structure that distinguishes Pakistan from most countries whose transport inventories are used as methodological references, but *less* than heavy goods vehicles and buses together (14.02 Mt). A model output, not a measured national finding, and one this edition changed.

This table is where correcting the allocation showed up, and the correction reversed its headline. The previous edition split each fuel control by vehicle-kilometres alone, which gives a motorcycle-kilometre the same petrol as a car-kilometre; on that basis two-wheelers took 42.3 % of on-road CO_2, 21.9 Mt, and the paper reported them as burning more than heavy goods vehicles and buses combined. Weighting each class's kilometres by its declared energy intensity moves them to 34.1 % and 13.40 Mt, and heavy diesel—trucks and buses at 14.02 Mt—becomes the larger claim on the sector. The sector total moved by 0.05 Mt in the process.

That is the third time in this paper's history that reading an input properly has overturned something it previously reported, after the ×16/9 control arithmetic and the OCAC tables it described as unobtained while holding them. The pattern is worth naming rather than quietly fixing: each correction landed in the

composition or the control provenance, never in the anchoring structure, which is the argument for the structure. It also cuts against the convenient reading of this sector—a two-wheeler-dominated fleet is a more novel finding than a truck-and-bus-dominated one, and it was the artefact.

The class table remains the sector's weakest output, and the corrected version is not better supported than the old one—it is less wrong in a knowable direction. Its three inputs are registration counts, assumed annual kilometres and declared intensity bands, none measured in Pakistan. What follows below prices that honestly rather than resting on the point estimate.

7.5. The fuel-based series the philosophy makes possible

The annual road-fuel series covers the same ten fiscal years as the national core, July 2015–June 2025. Fuel controls determine the total; vehicle classes provide a conditional partition. Monthly OCAC editions are held from July 2021. Earlier annual activity is allocated with explicitly weaker support and is not described as observed monthly use. Exhaust timing, evaporation and wear retain their own process evidence labels rather than inheriting one fuel profile universally.

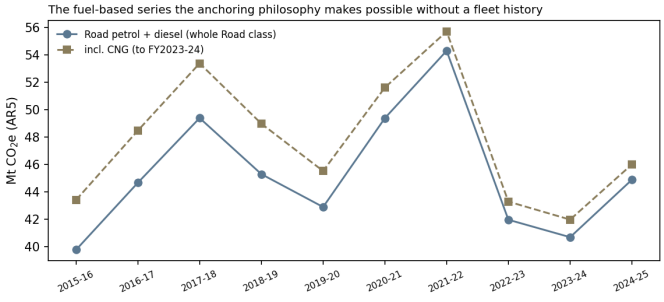


Figure 3: Road-class petroleum emissions, FY2015–16 to FY2024–25, with CNG added where the OGRA series reaches (`outputs/transport_fuel_series.csv`). Whole-Road-class basis, declared: no generator carve-out backcast, no off-road partition, no class composition. The 2022 peak and the post-2022 demand contraction are the feature EDGAR's own series shows and a fixed-year comparison would have misread.

The series is deliberately spartan—whole-Road-class, fuel-based, with a test asserting its reference-year row equals the inventory's petroleum total modulo exactly the two declared refinements—because composition over time would require the fleet history the sector does not have, and inventing it would spend the credibility the fuel basis earns.

7.6. Provincial allocation, now from fuel sales

The petroleum streams are allocated on provincial fuel sales—the monthly area pages of the same OCAC publication the national controls come from, summed and shared per fuel—replacing the registration proxy whose Sindh result this paper previously declared indefensible. The shape is the reference year's *own*: twelve FY2023–24 area pages, from OCAC's direct single-month editions (`data/activity/transport/ocac_sectoral_provincial.csv`).

An earlier edition of this paper borrowed FY2024–25's shape here, on the claim that OCAC printed complete

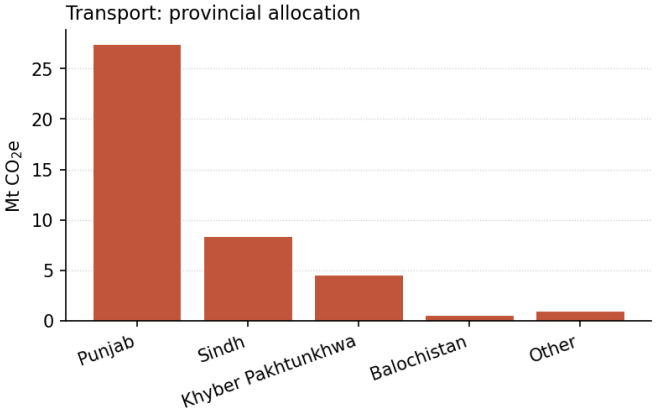


Figure 4: Punjab 27.49 Mt, Sindh 8.36 Mt, Khyber Pakhtunkhwa 4.52 Mt, Balochistan 0.58 Mt, other territories 0.98 Mt. The petroleum streams ride provincial fuel sales from OCAC's monthly area editions, per fuel; CNG and the off-road class ride their own proxies (Table 13). Fuel sold is still not fuel consumed (freight crosses provincial boundaries), which is the residual caveat.

area pages "for FY2024–25 and FY2021–22 but not for the reference period between them". That was true of the cumulative period-to-date editions the panel was then built from, which carry a national page only—and false of the publication: the direct editions of the same months carry full area pages, and the repository held all thirty-six of them when the sentence was written. The bounding argument offered for the borrowing was sound—the FY2021–22 bracket bounded it at 3.86 percentage points—and, measured now against the reference year's own pages, the borrowed shape was in fact off by at most 0.45 points on any (province, fuel) share. A sound bound over a false premise is still not the year's own data. The neighbouring panel years bracket the own-year shape in `outputs/transport_provincial_shape_brackets.csv`, where the largest drift against either neighbour is 3.93 points—the empirical answer, kept published, to whether borrowing a provincial shape across one year is ever defensible here. Not every stream rides fuel sales, and Table 13 says exactly which does:

Stream	National control	Provincial proxy
Road petrol	OCAC Road MS+HOBC	provincial petrol sales
Road diesel	OCAC Road HSD, net of off-road partition	provincial HSD sales
CNG	OGRA national metered gas	CNG fleet energy proxy
Mixed off-road	HSD partition	cropland proxy

Table 13: The allocation crosswalk: which stream rides which provincial proxy. CNG's geography still rides the fleet model (whose CNG kilometres the two-basis check flags as overstated), now weighted by the same declared energy intensities as the class split rather than by distance—three-wheelers are 43 % of CNG energy against 74 % of CNG distance, and the correction moves 4.7 points out of Punjab. The off-road class rides cropland, right for tractors and wrong for the residual's remainder.

The change is not cosmetic. Registration location gave Sindh 6.3 % of road activity, below Khyber Pakhtunkhwa, for the province containing Karachi and the port system; fuel sales give it 19.45 % of road diesel, and diesel is allocated

separately from petrol precisely because freight geography is not passenger geography.

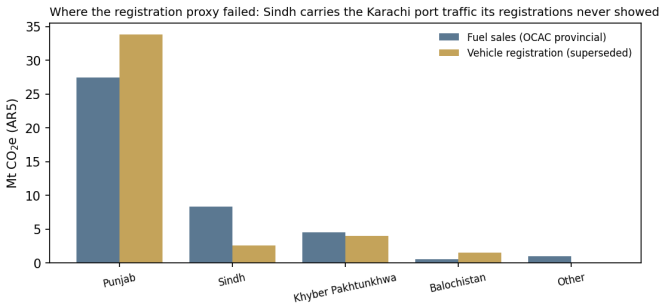


Figure 5: The failure diagnostic: the superseded registration allocation against the fuel-sales allocation (`outputs/transport_provincial_proxy_comparison_2024.csv`). Registration location understated Sindh by roughly a factor of three—vehicles are registered where it is administratively convenient and fuelled where they work.

The remaining caveat is stated rather than hidden: fuel sold in a province is not fuel consumed there, because freight transits. It is, however, the right error to be left with—a sales-side blur on a measured quantity, rather than an administrative register standing in for activity.

7.7. Contribution to the national inventory

Category	Level (%)	Variance (%)
Road transport	9.04	0.90
Mixed off-road (1A4c/1A2)	0.39	0.01

Table 14: Shares of the national total and of national Approach-1 variance. At IPCC category level road transportation is the second-largest category, marginally above public electricity generation; the transport module is the third largest category of calculation module, behind agriculture and power. It contributes little to the uncertainty on the national total.

That contrast follows directly from anchoring. Road transport is 9.0 % of Pakistan's emissions and 0.9 % of its inventory variance, a ratio of nearly ten to one in the direction of "large and parametrically well constrained under the current boundary".

These are COMMON-PERIOD shares, all on FY2023–24. Measured against an FY2021–22 transport year, the share would be 12.2 %; on FY2023–24 it is 9.0 %, and road transport remains the second-largest category, only marginally above public electricity generation.

That movement is a demand history, not a methods change. Petrol sold to the Road customer class fell from 9128 kt in FY2021–22 to 7209 kt in FY2023–24 and diesel from 7887 to 5551, so the sector was carrying a figure from the demand peak that preceded the post-2022 contraction while the sectors it was ranked against had moved on two years. The previous edition predicted this: it computed the matched-year share from the fuel series and reported it as 9.43 % without adopting it. The realised value is 9.02 %, and the small remaining difference is the two refinements the series deliberately omits—the generator carve-out and the off-road partition. The reason is structural: the activity datum is a metered, sold, sectorally partitioned fuel quantity with comparatively narrow bounds, and the emissions are almost entirely CO_2 from a well-characterised factor.

The practical implication is a useful negative result. Refining Pakistan's vehicle fleet model would improve the class composition—which matters for air quality and for projecting fleet policy—but would move the national inventory's uncertainty band hardly at all. Fleet modelling is where transport inventory effort in developing countries usually goes, and on this evidence it is not where the national total's uncertainty is.

8. Discussion

8.1. What anchoring buys, and what it costs

Anchoring gives the sector total three properties that a bottom-up estimate does not have. It agrees with the national fuel statistics exactly rather than approximately. It is insulated from the fleet model, so the least supported input in the sector cannot move the most used output. And it makes the boundary problem—which customer classes are road, and what a retail sales channel can and cannot see—visible as an executable ledger rather than absorbed into an unexamined assumption about what fraction of diesel is "road".

The cost is real and worth naming. The class composition in Section 7 is determined entirely by the fleet model, with no independent check, so a reader wanting emissions by vehicle type gets a number whose uncertainty this paper cannot quantify. A bottom-up inventory would produce the same weak composition but would additionally propagate it into the total; anchoring confines it. That is a better trade, not a free one.

There is also a subtler point, which Supplementary Note S1 quantifies. Removing the fleet uncertainty from the total leaves the shared fuel-carbon factor as the dominant term. An earlier edition read the cross-sector correlation scenario as roughly doubling the band; the common-random-numbers rebuild showed that spread to be a sampling artifact, and both treatments of the pooled Monte Carlo now print the same ±12.9 %, beside the sector's own Approach 1 interval of ±11.4 % that the results report as the number to quote. Anchoring does not make the sector more certain; it makes the remaining uncertainty legible.

8.2. What the re-anchor changed, and what it teaches

Reading the controls from their primary table moved the sector by six megatonnes and flipped the EDGAR comparison from above the benchmark to below it. Neither number was the point. The previous edition's control was arithmetically inconsistent with its own source note, and the matched-year benchmark agreement it produced was coincidence—the wrong annualisation happened to land near the right vintage. The lesson generalises: a benchmark comparison tests consistency, not provenance, and an inventory that treats agreement with EDGAR as validation has its evidentiary arrow backwards.

The re-anchor also shrank the boundary problem rather than merely renaming it. Bulk industrial, utility, government

and railway diesel left the road control by the structure of the source; what remains unresolved is the retail channel's end-use split, bounded by the ten-year road-share panel and carried as status rows. The residual ask of Pakistani statistics is now precise: not "publish a sectoral split"—OCAC does—but an end-use survey of the retail diesel channel, separating road vehicles from tube wells and captive gensets at the pump.

The asymmetry the previous edition claimed survives, strengthened: this inventory can name what is wrong with its own transport figure more precisely than any external comparison can. A ratio says two numbers differ; the ledger says which claimant is unresolved, in which channel, inside which row, within what bound.

8.3. Implications

Fuel controls and fleet assumptions constrain different outputs. Fuel anchoring constrains combustion CO_2 and isolates the effect of class partition on that total. PM, NO_x, evaporation and wear remain sensitive to vehicle technologies, controls, maintenance and operating conditions. A GHG variance screen therefore cannot establish the value of fleet evidence for the air-pollutant product.

Retail end-use evidence would constrain the source boundary. The unresolved fraction of retail diesel used by agricultural and captive equipment affects the road/off-road assignment directly. This evidence serves a different purpose from fleet technology and vehicle-kilometre information, which constrain pollutant factors and class composition. Their value cannot be ranked using fuel-derived CO_2 alone.

Two-wheelers are the largest single class, but not the largest claim. Under the reference allocation they receive 34.1 % of on-road CO_2; the superseded run's composition envelope spanned twelve points around it, and in every one of them heavy goods vehicles and buses together (14.02 Mt at the reference) exceed them. The previous edition, splitting fuel by distance alone, put two-wheelers at 42.3 % and above heavy diesel; that ordering was an artefact of the weighting and this edition retires it.

We draw no policy conclusion from the ordering, and the earlier draft's remark that car-focused electrification addresses only a minority of the sector is withdrawn—not because it is necessarily wrong, but because nothing here supports it. A ranking whose inputs are registration counts, assumed kilometres and declared intensities, none of them measured in Pakistan, is not a basis for allocating policy attention between vehicle classes. What the composition can support is a weaker and more useful statement: whichever end of the envelope holds, this is a fleet in which two-wheelers and heavy diesel each account for a substantial share, and an inventory that resolved only one of them would mislead.

9. Limitations

The class split has no independent validation and should be read as the sector's weakest output. No nationally representative Pakistani vehicle-kilometre survey has been identified for this study, and no on-road fuel-economy measurement either. The shares rest on three inputs, none of them measured in the country: registration counts, assumed annual kilometres, and the declared per-kilometre energy intensities that weight them. The largest single result—two-wheelers at 34.1 % of on-road CO_2 under the reference allocation—rests on the least surveyed class in the fleet and on a judgement about how much fuel a motorcycle burns.

This edition changed that result, which is itself a limitation worth stating: the previous allocation weighted kilometres alone and put two-wheelers at 42.3 %, above heavy goods vehicles and buses combined. Correcting the weighting reversed that ordering. A reader should conclude that the composition is not yet settled, not that it is now settled differently. Anchoring confines the consequence to composition rather than total, which is the reason the method was chosen, but it does not make the composition reliable.

The off-road class is a residual as well as a category, and part of it belongs to a different IPCC code. The source workbook maps its catch-all "other vehicles" entry into off-road wholesale, and the whole class is then modelled as 100 % diesel. The magnitude of the carve-out is therefore a genuine partition of the fuel total, while what is inside it is soft. Whatever construction and industrial machinery the residual contains belongs under 1A2 gvii, not 1A4c; this inventory reports the whole class under the mixture-declaring name *Mixed off-road (1A4c/1A2)*, declares the misassignment in the completeness table, and prices the reporting recode as a scenario, because splitting the residual needs equipment stock data no source here provides. The cropland proxy that allocates it provincially is right for tractors and wrong for whatever else the residual contains.

Retail-bought agricultural and captive diesel remain inside the road residual. Tube wells, threshers and captive generators fuelled at retail pumps draw on the Road customer class, this inventory has not separated them, and they bias the road residual high—conditional on the other deductions, since errors in the generator carve-out or the off-road partition could run either way. The claim is framed rather than open-ended: bulk agricultural HSD is observed zero in all ten OCAC editions; the ten-year road share of national diesel (0.83–0.91) is a customer-class boundary sensitivity that bounds the sales channel, not the contamination within it; and the retail non-road scenarios price the within-channel claim at 2.5–10% of the Road row (Table 10). Resolving it needs a retail-channel end-use survey, which no identified source provides. The ledger carries three claimants as UNRESOLVED status rows—the third, "other retail stationary/non-road uses", because the named two are not proven exhaustive.

The provincial allocation now rides fuel sales, whose residual blur is transit. Fuel sold in a province is not fuel consumed there—freight crosses provincial boundaries, so Sindh's 19.45 % of road diesel partly reflects fuel bought at Karachi and burned en route north. The shape is the reference year's own twelve area pages—an earlier edition borrowed FY2024–25's here on the false premise that none were published for the reference period, a borrowing since measured at no more than 0.45 percentage points of error on any share.

This is a far smaller distortion than the superseded registration proxy's (which understated Sindh threefold; the comparison is published as a diagnostic), but it is a real one. The release now ships the road geometry that a finer allocation would start from—`outputs/pakemis_road_length_adm2.csv` and a 0.05° major-road density raster, extracted from a dated, hashed OSM snapshot—as a diagnostic only: road length is not traffic, and moving fuel onto the network would need traffic observations this inventory does not hold. CNG's provincial split still rides the fleet model and the off-road class rides cropland (Table 13); only the petroleum streams ride fuel sales.

The CNG heating value is unpinned. CNG is estimated from OGRA's metered sector gas, but OGRA reports a volume and IPCC works in net terajoules: the conversion applies a gross heating value and then the 0.90 gross-to-net factor for natural gas. The gross-to-net step is a documented convention; the gross value is not—Pakistani pipeline gas runs 900 to 1000 Btu/scf depending on field and RLNG share and nothing read here pins it. Both ends are reported; the span is 0.13 Mt, a tenth of the CNG stream. This is the same unresolved quantity as the captive-power category's heating value, and the two sectors are pinned to the same values by test so that one inventory cannot assume the same gas two ways.

The fleet model overstates CNG travel, and this sector has evidence rather than a suspicion. Section 6 divides the metered gas by the modelled CNG vehicle-kilometres and gets 0.91 MJ km^{-1}, below any plausible value for a fleet seven-tenths three-wheelers by distance. The consequence is confined to what rides vehicle-kilometres—wear, fleet composition and the CNG streams' provincial split—because the CNG emissions come from metered gas. It is nonetheless the clearest evidence in this paper for the sector's central methodological choice.

Air factors are traceable transfers, with unresolved fleet applicability. The current EMEP/EEA factors are linked to held source tables. Exhaust (Tier 1 factors averaging hot and cold operation for a typical 2010 EU-27 fleet, with no Pakistan-specific cold-start treatment), petrol evaporation, tyre/brake wear and road-surface wear are included. Resuspended road dust remains unestimated. The fleet composition and CNG-to-LPG and petrol-bus substitutions remain conditional. Sulfur stoichiometry uses declared annual policy/specification cases rather than measured universal compliance. Correct fuel closure does not establish the PM or NO_x factor's applicability.

The time series is fuel-only, by design. The ten-year fuel series (Section 7.5) carries no generator back-cast, no off-road partition and no class composition, because the inputs those refinements need exist only for the reference year. The declared whole-Road-class basis keeps the series honest; it also means the series is a magnitude record, not a structural one.

Emission factors are Tier 1 defaults throughout. They carry assumptions about technology mix and combustion conditions derived largely from countries whose vehicle fleets differ substantially from Pakistan's—most consequentially for the two-wheeler class, which dominates here and is marginal in the fleets the defaults were derived from.

The controls are member-supplied. OCAC's own disclaimer states its data are provided by members and not independently verified. The declared 10 % control uncertainty is anchored on the independent cross-check available—the railway figures, 3.8 % apart year-matched and 11.2 % on the conservative cross-year pairing—and the panel's internal closures, and it is a judgement, not a fit.

10. Conclusion

Pakistan's road transport—the conditional road residual under the current claimant assignment—emitted 40.19 Mt CO_2e in FY2023–24 and its mixed off-road machinery 1.73 Mt, together 9.4 % of a ten-sector national inventory in which all ten sectors now report the same twelve months. Both are anchored on the road customer class of the national sectoral sales table, plus metered transport CNG, so the sector total agrees with Pakistan's fuel statistics by construction rather than by adjustment.

Three results generalise beyond this country.

Anchoring puts the weakest input where it does least damage. A fleet model that distributes a known quantity of fuel cannot move the sector total; one that sizes it can. The class composition reported here is no better supported than any bottom-up equivalent—worse, in a country with no identified vehicle-kilometre survey—but it is confined to the output where its weakness matters least. The arrangement is what the IPCC guidance implies where fuel statistics are strong and fleet data weak, and it is rarely stated as a design choice.

A customer-class table is a claimant ledger waiting to be read. Pakistan's sectoral sales statistics partition national fuel exhaustively—road, rail, navigation, power, industry, government—and reading the partition into a two-tier ledger replaced most of a boundary argument with observation: bulk claimants excluded by the source's own structure, one carve-out relayed from the sector that measures it, one partition, an independent cross-check where two unrelated sources report the same boundary row, and the genuinely unresolved remainder—the retail channel's end-use split—carried as priced status rows, because naming a residual's direction and sensitivity is more useful than a correction nobody can evidence.

A benchmark match is not provenance. The previous edition's control total was arithmetically inconsistent with its own source note, and a matched-year EDGAR comparison failed to notice, because the wrong number landed near the right vintage. Correcting the provenance moved the sector six megatonnes and flipped the comparison's direction; on a scope inferred from EDGAR's category-resolved source files and a fiscal-year-blended benchmark on the reference period the estimate now sits at 0.85 of the blend—a scope-approximate diagnostic, with the rc4-era 0.70 retained in the comparison table as a superseded mixed-vintage construction. The paper retains the superseded control as a named executable scenario—the bridge readers of the earlier edition are owed—and draws the general moral: external agreement tests consistency, and only provenance tests provenance. The parallel-table N_2O trap (Section 5) makes the same point from

the factor side: a CO_2-dominated validation cannot detect a category running 8 % low through the wrong trace-gas table.

Fuel controls constrain the GHG magnitude, while end-use claimant evidence constrains the road/off-road boundary. Fleet technologies and operation remain consequential for pollutant factors even when class shares have little effect on fuel-derived CO_2. The annual national core covers July 2015–June 2025; the detailed reference-year class analysis does not constitute observed historical fleet composition.

Archived input tables, source-factor registers, build commands, plotted data and numerical checks accompany the sector estimates. The release distinguishes regeneration from compiled tables from a full primary-source extraction.

Code and data availability

The calculation engine, the activity data, the fleet model and every output quoted here are version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors); an archived release with a Zenodo DOI is minted at submission. The sector is regenerated with `python3 -m pkinv.sectors.transport.run_transport`, which requires the railways and residential modules to be importable for the boundary relays; Appendix A gives the full sequence, and the run manifest hashes every primary source. Not every input source is independently published: the archived input package ships the internal Punjab transport manuscript and cached source extracts, and the per-kilometre air factors carry no located primary citation and are labelled as such wherever they appear. Every number printed in this paper is read from a generated CSV rather than transcribed, and a test asserts that the manuscript and the engine agree.

References

Ahmad, R., 2026. PakEmis 1.0.0rc68: greenhouse gas and air pollutant emissions in pakistan, july 2015–june 2025. National data-description manuscript with technical report and sector notes; local review package prepared. External repository review deposit and persistent identifier pending; no journal submission asserted.

Climate TRACE, 2026. Climate TRACE emissions inventory and data downloads. https://climatetrace.org/data. Version 7 country- and source-level API records; CC BY 4.0; accessed 6 August 2026.

Crippa, M., Guizzardi, D., et al., 2025. GHG emissions of all world countries — 2025 report (EDGAR GHG booklet). European Commission, Joint Research Centre. Sheet GHG_by_sector_and_country; annual series 1970–2024; CC BY 4.0. Snapshot archived at `sources/EDGAR/EDGAR_2025_GHG_booklet_2025.xlsx`, sha256 `8bd6b8a4...` pinned per benchmark row in `data/benchmarks.csv`. https://edgar.jrc.ec.europa.eu/.

Crippa, M., Solazzo, E., Huang, G., Guizzardi, D., Koffi, E., Muntean, M., Schieberle, C., Friedrich, R., Janssens-Maenhout, G., 2020. High resolution temporal profiles in the Emissions Database for Global Atmospheric Research. Scientific Data 7, 121. doi:10.1038/s41597-020-0462-2.

Government of Pakistan, 2025. Fortnightly administered retail prices of petroleum products (motor spirit), gazette notifications with OGRA computations. Band over July 2024–June 2025 approximately Rs 249–293 per litre; no annual average is published.

Guttikunda, S.K., 2024. Vehicle stock numbers and survival functions for on-road exhaust emissions analysis in India: 1993–2018. Sustainability 16, 6298. doi:10.3390/su16156298.

IPCC, 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2: Energy, Chapter 3: Mobile Combustion. Technical Report. Institute for Global Environmental Strategies, Japan. URL: https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html.

Ministry of Climate Change, Government of Pakistan, 2022. Pakistan's First Biennial Update Report to the United Nations Framework Convention on Climate Change. Technical Report. Ministry of Climate Change, Islamabad. URL: https://unfccc.int/documents/. national greenhouse-gas inventory for FY2017–18; submitted to the UNFCCC.

Ministry of Finance, 2025. Pakistan Economic Survey 2024–25, Chapter 14: Energy. Technical Report. Government of Pakistan. Figure 14.4, petroleum product sales by product (source: Oil Companies Advisory Council).

Naveed, M., 2024. District-level road transport emission inventory for punjab province, pakistan: A 2015–2023 multi-year assessment. Internal manuscript, Pakistan Air Quality Initiative, Lahore (undated; year inferred from its 2015–2023 coverage and 2022–23 source citations). Distributed with the archived input package as sources/Punjab Transport/Punjab Transport.pdf.

Oil and Gas Regulatory Authority, 2024. State of the Regulated Petroleum Industry 2023–24. Technical Report. Government of Pakistan. Islamabad. URL: https://www.ogra.org.pk/annual-reports.

Oil and Gas Regulatory Authority, 2026. Petroleum industry reports archive. Official report archive. URL: https://www.ogra.org.pk/ogra-annual-reports. accessed 7 August 2026.

Oil Companies Advisory Council, 2026. Oil industry statistics archive. https://www.ocac.org.pk/oil-industry-statistics/archives/. Annual sectoral-sales reports from FY2016–17 and monthly reports; accessed 6 August 2026.

Pakistan Bureau of Statistics, 2025. Household Integrated Economic Survey (HIES) 2024–25: Microdata and Key Findings Report. Technical Report. Pakistan Bureau of Statistics, Government of Pakistan. URL: https://www.pbs.gov.pk/content/microdata. first digitally enumerated round; 32,814 sampled households, 30,123 completed. SPSS microdata files; consumption module Section 6A.

Pakistan Railways, 2024. Year Book 2023–24. Technical Report. Ministry of Railways, Government of Pakistan. Fuel consumption tables; cached text extracts ship with the repository under sources/Railways/.

Provincial Bureaux of Statistics, 2024. Registered motor vehicles by type and district, provincial statistical tables (punjab, sindh, khyber pakhtunkhwa, balochistan). Provincial development statistics editions, various years. Cumulative registered stock; transcribed to data/activity/transport/vehicles_district.csv with the source workbook's citation sheet.

Sui Northern Gas Pipelines Limited, 2024. Annual Report 2024: Financial Year 2023–24. Technical Report. Sui Northern Gas Pipelines Limited. URL: https://www.sngpl.com.pk/download/AnnualReport-2024/SNGPLAR2024.pdf.

Sui Southern Gas Company Limited, 2024. Annual Report 2024. Technical Report. Sui Southern Gas Company Limited. URL: https://www.ssgc.com.pk/web/wp-content/uploads/pdfs/2024_annual_report.pdf.

Appendix A. Reproduction

The sector runs on Python 3.11 with no compiled dependencies:

```
pip install -e .
python3 -m pkinv.sectors.transport.run_transport
python3 -m pkinv.sectors.transport.air_pollutants
```

which writes `transport_totals_2024.csv`, `transport_by_stream_2024.csv`, `transport_fuel_2024.csv`, `transport_provincial_2024.csv`, `transport_air_pollutants_2024.csv`, the twelve-month layer (`transport_monthly_2024.csv`, each month's share with the emissions it implies and the basis of the shape), the two-tier fuel-claimant ledger (`transport_fuel_ledger_2024.csv`), the uncertainty system (`transport_uncertainty_2024.csv` and its variance-decomposition, inputs-register and structural-scenario companions), the ten-year fuel series (`transport_fuel_series.csv`), the provincial proxy comparison (`transport_provincial_proxy_comparison_2024.csv`), the mass-to-energy fuel bridge (`transport_fuel_bridge_2024.csv`), the OCAC consistency diagnostic (`transport_ocac_control_diagnostic.csv`) and a run manifest (`transport_run_manifest_2024.json`) that hashes every primary source—including all 57 OCAC sectoral PDFs—to `outputs/`.

The OCAC sectoral CSVs the controls read are committed data; regenerating them from the PDFs is a separate, deterministic step (`python3 scripts/ingest_ocac_sectoral.py`, requiring `pdftotext`), and an ingest test pins the committed cells to independently verified values.

The relays are not constants in this sector. Household-generator petrol is read from the residential module (a carve-out) and railway diesel from the railways module (a cross-check), so reproducing the boundary requires both to be importable:

```
python3 -c "\
from pkinv.sectors.residential import combustion
print(combustion.generator_petrol_kt())"
python3 -c "\
from pkinv.sectors.railways import traction
from pkinv.sectors.railways import data as rw
print(traction.diesel_kt(rw.RailwaysData()))"
```

Figures are regenerated with `python3 -m pkinv.report.figures` and copied into this manuscript by `python3 scripts/sync_figures.py`; a test fails if any shipped figure differs from the output it plots. The full suite runs with `python3 -m unittest discover tests`.

Appendix B. Sources

Fuel controls are the Road customer-class rows of OCAC's Sectoral Sales of Energy Products, annual and monthly-provincial editions (Oil Companies Advisory Council, 2026), with the superseded Economic Survey-derived pair (Ministry of Finance, 2025) retained as a scenario; transport CNG is OGRA's sectoral gas statistics (Oil and Gas Regulatory Authority, 2024, 2026), with the billing heating-value basis from the two gas utilities (Sui Northern Gas Pipelines Limited, 2024; Sui Southern Gas Company Limited, 2024). Vehicle counts by class and province come from the provincial registration tables (Provincial Bureaux of Statistics, 2024). The Weibull survival parameters and the per-class MJ/km intensity bands are in `data/transport_params.csv` with a source string on every row—the survival parameters from the Punjab district-level transport manuscript (Naveed, 2024), which adopts Guttikunda (2024); the intensity bands are this inventory's own declared engineering judgement and say so. The annual kilometres and fuel splits are separate tables (`data/activity/transport/vkt.csv`, `fuel_split.csv`), transcribed from the same manuscript's workbook, which carries no per-row citation for them; they are declared without primary provenance in their own source columns and registered as a gap (`transport-class-activity-provenance`). A previous edition of this appendix said all three lived in the parameter table with sources recorded per row, which was not true of two of them. The rail cross-check reads the Pakistan Railways Year Book (Pakistan Railways, 2024) through the railways module; the generator carve-out reads HIES (Pakistan Bureau of Statistics, 2025) over the administered petrol price (Government of Pakistan, 2025)

through the residential module. Emission factors are IPCC 2006 Guidelines Volume 2 Chapter 3 defaults (IPCC, 2006): Table 3.2.2 for road, Table 3.3.1 for off-road machinery. Air factors come from held EMEP/EEA 2023 source chapters, with separate fuel, evaporation and wear units and an explicit class/fuel bridge (Section Appendix D). Supplementary Note S2 renders the current source tables and clearly labels the superseded diagnostic rows as historical. External comparison is EDGAR's series (Crippa et al., 2025), read at this sector's fiscal year as a blend of the 2023 and 2024 columns (48.13 Mt); the CY2021/CY2022 blend survives only as the superseded rc4 construction in the comparison table.

Every factor in the repository carries a source string, and a provenance test asserts that none is empty and that any informally sourced value is declared as such. The declared MJ/km intensity bands remain engineering assumptions rather than Pakistani operating measurements. Source-based air factors likewise require an applicability assessment for the transferred fleet and fuel technology. A source string is not the same thing as a citation, and the distinction is made explicit rather than averaged away.

Appendix C. Supplementary Note: why the shares are normalised over all classes

The off-road class is included in the denominator that normalises the vehicle-kilometre shares, and removing it is a mistake worth documenting because it is the natural thing to try.

Suppose off-road were dropped from the fleet before normalising, on the reasoning that this sector reports road transport. The remaining classes' shares would then sum to one over the *whole* diesel supply, so on-road vehicles would be assigned 100 % of the national diesel. Any separate estimate of off-road machinery—whether made here, in another sector, or by another author—would then claim its fuel a second time. The national diesel figure would support more emissions than there is diesel.

Keeping off-road inside the normalisation makes the two categories an exhaustive partition of one fuel quantity: their shares sum to one, so their emissions sum to the control total times the factor. The memo item that formerly carried off-road as a CO_2-only note outside all totals is asserted to be empty, which is the test that catches a future edit reintroducing the double count.

The same structure appears in this inventory's waste sector, where one municipal waste generation figure is partitioned between disposal and open burning (Ahmad, 2026). In both cases the useful property is not that the split is right, but that an argument about the split cannot change how much of the underlying quantity exists.

Appendix D. Supplementary Note: Road-Transport Air Pollutants (EMEP/EEA Tier 1)

This note reports the FY2023–24 road air-pollutant estimate using the EMEP/EEA 2023 Tier 1 factors. Native pollutant masses remain separate from greenhouse-gas aggregates. The national release extends the same source methods through FY2024–25 with source-year activities and declared factor applicability.

Fuel is the activity, and it reconciles to the controls by construction. Tier 1 exhaust factors are grams per kilogram of fuel; the fuel each vehicle class burns is the OCAC Road control total for the year—the same panel the greenhouse-gas side burns, net of the same household-generator carve-out—split between classes by the fleet model's energy-weighted class shares. Modelled fuel therefore equals sold fuel for every fuel in every year, the anchor the superseded diagnostic explicitly lacked; the class *split* still rides the fleet model, so composition remains the sector's weakest axis while the fuel quantity is constrained. Four process families are estimated as separate atoms: hot-plus-cold exhaust (fuel-based), evaporative petrol NMVOC (per-vehicle-day, at the guidebook's 20–35 °C daily temperature range, on the fleet model's own active vehicle counts), and tyre-and-brake and road-surface wear (per vehicle-kilometre). Black carbon is the guidebook's f-BC fraction of exhaust $PM_{2.5}$—a component of particulate mass, never additional—and SO_2 is sulfur stoichiometry under the declared per-year fuel-sulfur regime of `data/activity/transport/fuel_sulfur_by_year.csv` (diesel on the same cited Euro-II/Euro-V mandate boundaries the railways layer burns). Where the guidebook publishes no Tier 1 factor for a Pakistani cell, the bridge table (`data/activity/transport/emep_class_map.csv`) declares a named proxy—light CNG vehicles take the passenger-car LPG chemistry, petrol buses the light-commercial petrol row—never a silent one.

Pollutant	Tier 1 (kt/yr)
CO	1430.4
NMVOC	535.1
NO_x	172.1
SO_2	12.2
$PM_{2.5}$	8.2
Black carbon	2.0
NH_3	2.6

Table 15: Road transport, FY2023–24, EMEP/EEA Tier 1 on OCAC-reconciled fuel, all processes. NMVOC is non-methane VOC; NO_x is NO_2-equivalent mass, the inventory-wide basis. $PM_{2.5}$ here is the $PM_{2.5}$ core (exhaust $PM_{2.5}$ is $PM_{2.5}$ by the chapter's own statement, plus the wear processes' $PM_{2.5}$); the wear processes' coarser TSP and PM_{10} strata are supplemental beside it, contained not additive, and appear in the factor appendix. The native factor definitions remain documented by process.

Appendix D.1. Interpretation and conditional uncertainty

A fuel balance constrains the amount consumed, but does not validate a transferred PM or NO_x factor. Tier 1 fleet averages describe the guidebook's reference technologies and operating conditions; Pakistan's vehicle ages, two-wheelers, maintenance, catalysts and duty cycles remain unresolved. The source min–max factor spans and explicit technology cases therefore describe consequential assumptions rather than observed national fleet confidence limits. CNG-to-LPG chemistry and the petrol-bus class substitution remain named

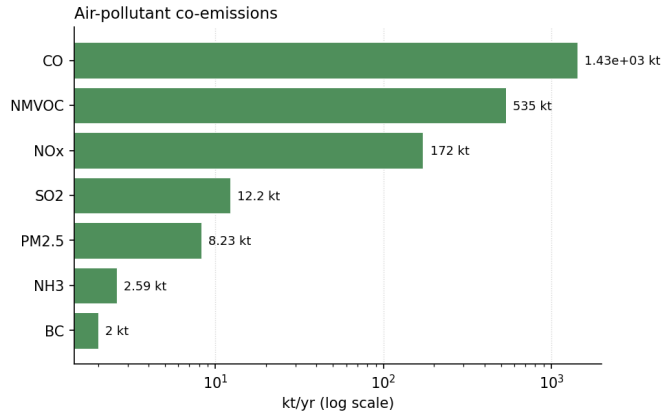


Figure 6: The Tier 1 layer, log axis: carbon monoxide exceeds black carbon by nearly three orders of magnitude, which a linear axis cannot show. Wear TSP and PM_{10} are excluded from the bars for the same containment reason the residential figure excludes them.

proxies. The conditional parameter ensemble retains its declared factor widths; separate technology and sulfur endpoints receive no assigned probability.

Exhaust, evaporation and wear retain distinct process records. Fuel-sales months support exhaust timing only as a combustion proxy. Evaporation and wear lack own-year monthly process evidence and use explicitly weaker allocation. Sulfur policy dates establish the specification scenario, not universal pump-level compliance. Primary particulate mass is an emission quantity; it does not provide ambient concentration, exposure or a health ranking.

Appendix D.2. Limitations of this layer

Tier 1 resolves fuel and broad vehicle category, not technology vintage, catalyst penetration or after-treatment; the class split rides the fleet model's declared intensity weights; the fuel-sulfur regime is declared from the import-mandate record, not measured at the pump, and domestic-refinery waivers mean the post-2021 central deliberately stays at the conservative pre-Euro-V specification; evaporative factors assume the guidebook's hottest daily temperature band year-round; road-dust resuspension stays out until a defensible local silt basis exists—a different claim, recorded in the gap register to stop it riding in.